

Gradient Boosting: From First Principles

Gradient Boosting builds models sequentially, where each new model tries to correct the mistakes of the current ensemble.

Why Gradient Boosting Is Needed

Some models underfit because they are too simple. Gradient Boosting improves prediction step by step by learning from residual errors.

Residuals

For regression, a residual is the difference between the actual value and the current prediction. Gradient Boosting trains the next tree to predict these residuals.

Core Idea

The model is built additively:

$$F_m(x) = F_{m-1}(x) + \eta h_m(x)$$

Here, $F_m(x)$ is the updated model, $h_m(x)$ is the new weak learner, and η is the learning rate.

Learning Rate

The learning rate controls how much each new tree contributes. Smaller learning rates often need more trees but can generalize better.

Classification

For classification, Gradient Boosting corrects mistakes using gradients of a loss function rather than simple residuals.

Limitations

Gradient Boosting can be very powerful but sensitive to hyperparameters, overfitting, and noisy data.